



Tacos in space

Astronaut makes zero-g tacos with 1st chile peppers grown in space.
<https://digit.in/nov21-11>



Lava on the moon!

Lunar rock samples from China's Chang'e 5 mission reveal that lava once flowed on the Moon. <https://digit.in/nov21-12>



REVIVING INTEGRAL

Bringing a satellite back from the dead

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he Integral (International gamma ray astrophysics laboratory) is a dedicated space

telescope for observing the most energetic events in the universe. The mission is led by the ESA, launched by a Russian Proton Rocket, with control support from the Goldstone Observatory, NASA's deep space network (DSN) ground station in the Mojave desert. The massive four ton satellite was deployed into an elliptical geosynchronous orbit in 2002, and has been providing us with gamma ray observations of specific targets for over 18 years. During each orbit, the satellite is hidden from the Sun for anywhere between four to six hours, which is known as the eclipse. During this

time, the solar arrays on the satellite cannot recharge using the Sun. The AntiCoincidence Shield or ACS on board, detects high energy particles no matter which direction they strike the instrument, and screens out everything that is not from the direction in which the telescope is being pointed. The ACS, thus acts like a full sky detector of gamma rays, and is very good at detecting sudden, unpredictable, short lived gamma ray bursts, a mysterious and little understood phenomenon. Within a few seconds, as much energy is released as the Sun can over its entire 10 billion year lifespan. Integral works by monitoring certain points of the sky for hours at a time, and during the observation run, has to make subtle but precise changes in its attitude, or angle at which it is pointing to its target, in order to eliminate background sources and noise. For its precise pointing operations, Integral uses a star tracker to orient itself, a system that uses the known and fixed positions of the stars. While Integral is

observing a target, it has to make sure that it is always pointed at the target. However, the Sun exerts pressure on the solar panels, making it tilt towards one side, and making the satellite slowly rotate. To counteract this extra force from the Sun, Integral uses three reaction wheels in three different axis to absorb the extra force. These wheels build up energy and reach a saturation point after which they can no longer absorb any extra momentum, at which point the satellite dumps the excess energy stored in the wheels, which took place every two to three days. The thrusters fire during the momentum dumping, to keep the satellite oriented.

In July 2020, Integral suddenly went into safe mode, where only the critical systems are running, and the solar panels orient towards the Sun, in what is considered to be a low risk orbit for the satellite. This itself was puzzling for the mission control, but additionally, the satellite appeared to be going through an eclipse, even though there wasn't one scheduled. Turns out that the satellite had turned all the way around, and was facing the Earth instead of the Sun, appearing to be in an eclipse. Turns out that the thruster had failed, the satellite had run out of fuel, and now was close to being junk in space. ESA researchers put their heads together, and came up with an innovative solution. Instead of counteracting, storing and dumping the pressure from solar radiation the researchers could actually use the solar pressure as an energy source. The approach was a carefully choreographed ballet where energy was sequentially stored and transferred between the reaction wheels, in a complicated series of steps known as the z-flip. No space agency had attempted such a maneuver before, and it worked, even without the critical thrusters to help the team. The satellite could then use the stored energy in the reaction wheels for attitude control, that is pointing in the right